

TEREZA SÁSKOVÁ

SUMMARY

Data Science MSc student at the University of Trento, working across machine learning, computer vision and data engineering. Recent work includes a multi-city ML pipeline for single tree detection from aerial imagery (MSc thesis with FBK), a Kafka/Spark recommender system, and a PostgreSQL booking system running in production. Seeking data science or ML roles.

WORK

Data Science Intern: FBK Digital Commons Lab, Trento · Feb – Aug 2026, continued as MSc thesis

- Built an end-to-end pipeline measuring the 3-30-300 urban-greenery rule
- Single tree detection without training labels, combining a domain-specific detection with a foundation model (DeepForest(RestinaNet + SAM2)

Data Systems Coordinator, Sport Šumava (2023–2025)

- Designed and built a production booking system (Flask, PostgreSQL, Railway)
- Cleaned and managed customer data

EDUCATION

Master of Science in Data Science
Università degli Studi di Trento, Italy
Relevant Coursework: Statistical Learning, Machine Learning, Big Data Technologies, High performance computing, Foundation models, Geospacial analysis, Signal, Image, Video

PROJECTS

Podcast Recommendation Platform
Hybrid recommender combining implicit-feedback ALS on streamed user events with transcript embeddings. Kafka, Spark, Delta Lake, Airflow, MongoDB.

Weekend Registration System: Ski Zadov · live: vikendovelyzovani.zadov.cz
Booking app where PostgreSQL enforces season quotas, weekly cutoffs and sibling accounts; advisory locks prevent double bookings. Python, Flask, PostgreSQL, Railway.

MTB Trail Centre Site Selection, Šumava · live map
Analysed 7 years of GPS ride data (3,045 rides, ~63,000 km) to find the best trail-centre location in a national park: grid-based trail network, global and local Moran's I, DBSCAN candidate sites, and suitability scoring under protected-zone constraints.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, C
Databases: DuckDB, PostgreSQL, MongoDB
Geospatial & Remote Sensing: GEE, QGIS, GDAL, OSMnx, Sentinel imagery processing, rasterio
Data Engineering & Big Data: Apache Spark, Kafka, Docker, Delta Lake
Machine Learning & Computer Vision: PyTorch, Scikit-learn, Pandas, NumPy, deep learning-based object detection/segmentation
High-Performance Computing: MPI, OpenMP, GPU-accelerated training/inference (cloud compute)